Visualize the data using Python libraries matplotlib, seaborn by plotting the graphs on Heart Disease Dataset

```
# import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Reading csv file
df = pd.read_csv('heart.csv')

df.head()
```

```
   age  sex  cp  trestbps  chol  fbs  restecg  thalach  exang  oldpeak
slope  \
0   52    1   0       125   212    0        1      168      0      1.0
2
1   53    1   0       140   203    1        0      155      1      3.1
0
2   70    1   0       145   174    0        1      125      1      2.6
0
3   61    1   0       148   203    0        1      161      0      0.0
2
4   62    0   0       138   294    1        1      106      0      1.9
1

   ca  thal  target
0   2     3       0
1   0     3       0
2   0     3       0
3   1     3       0
4   3     2       0
```

```
# Check info
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1025 entries, 0 to 1024
Data columns (total 14 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   age       1025 non-null   int64
 1   sex       1025 non-null   int64
 2   cp        1025 non-null   int64
 3   trestbps  1025 non-null   int64
 4   chol      1025 non-null   int64
```

```
 5   fbs       1025 non-null   int64
 6   restecg   1025 non-null   int64
 7   thalach   1025 non-null   int64
 8   exang     1025 non-null   int64
 9   oldpeak   1025 non-null   float64
 10  slope     1025 non-null   int64
 11  ca        1025 non-null   int64
 12  thal      1025 non-null   int64
 13  target    1025 non-null   int64
dtypes: float64(1), int64(13)
memory usage: 112.2 KB
```

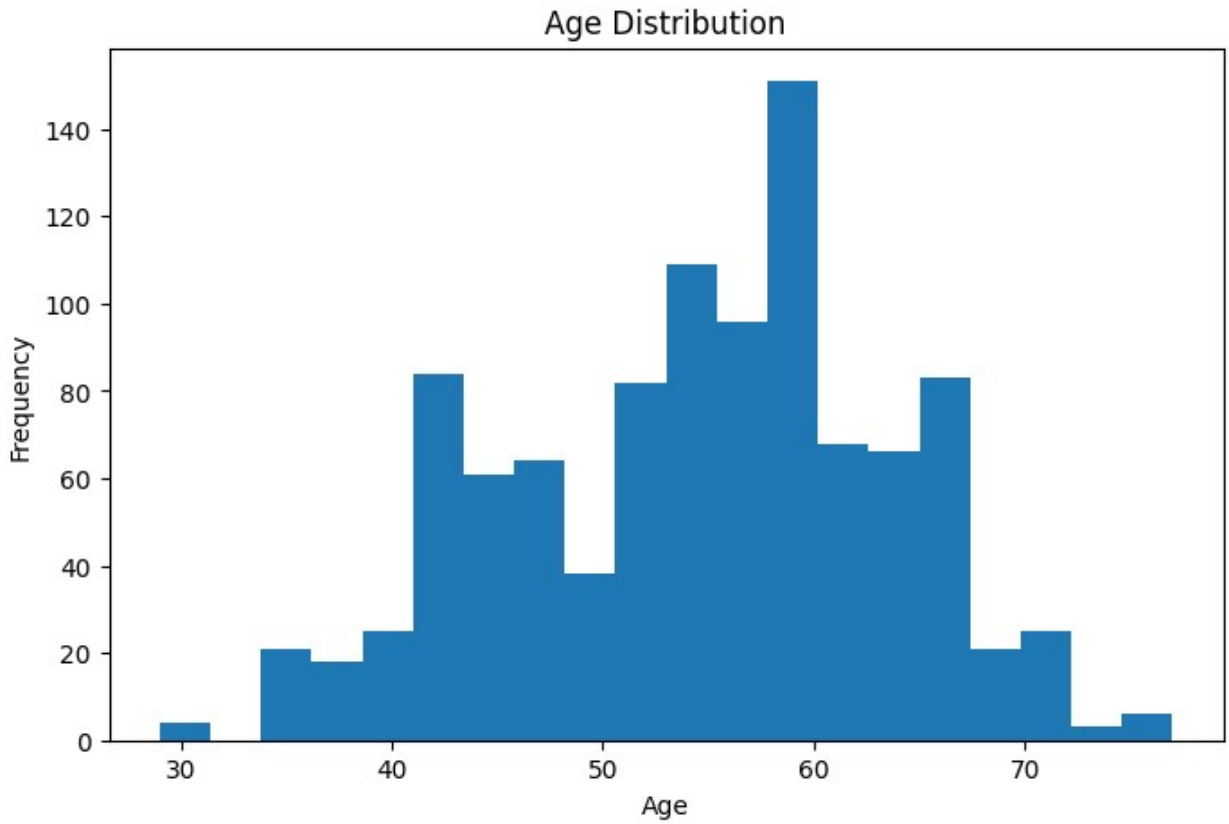
```
# Check missing values
df.isnull().sum()
```

```
age         0
sex         0
cp          0
trestbps    0
chol        0
fbs         0
restecg     0
thalach     0
exang       0
oldpeak     0
slope       0
ca          0
thal        0
target      0
dtype: int64
```

```
# Rename target column if needed
# 1 = disease, 0 = no disease
df['target'] = df['target'].astype('category')
```

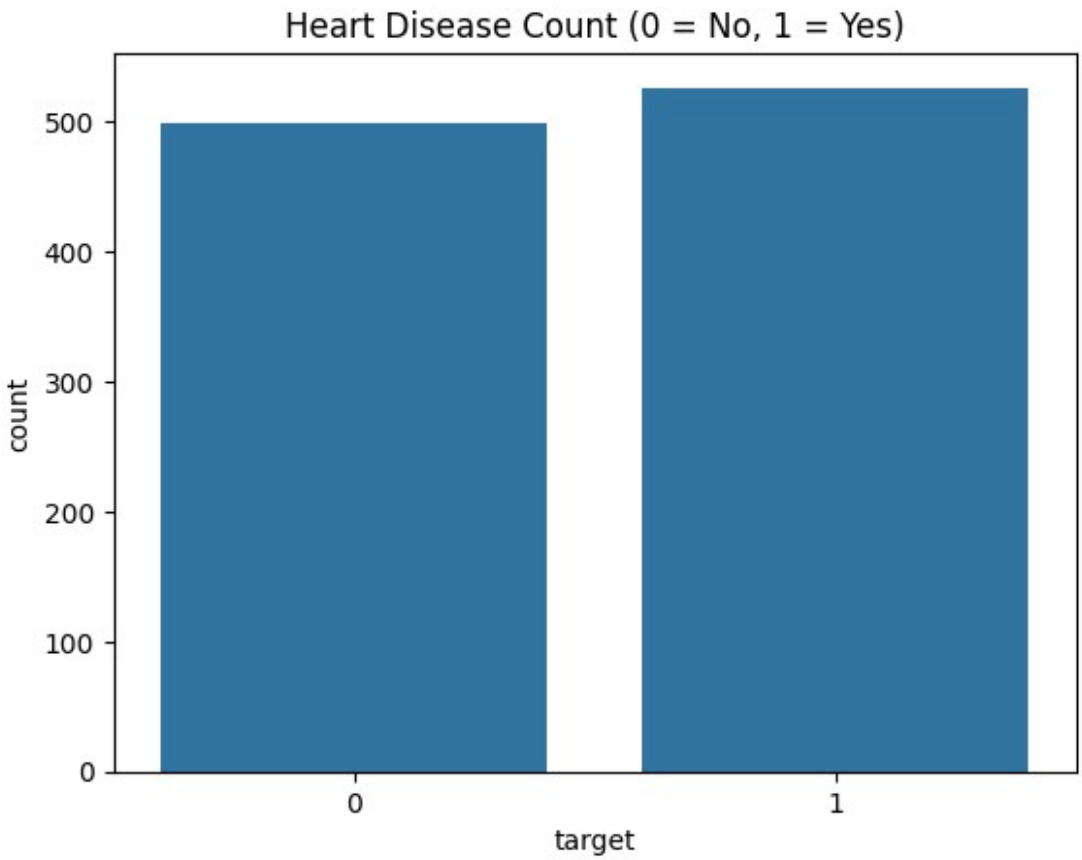
Histogram

```
plt.figure(figsize=(8,5))
plt.hist(df['age'], bins=20)
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.show()
```



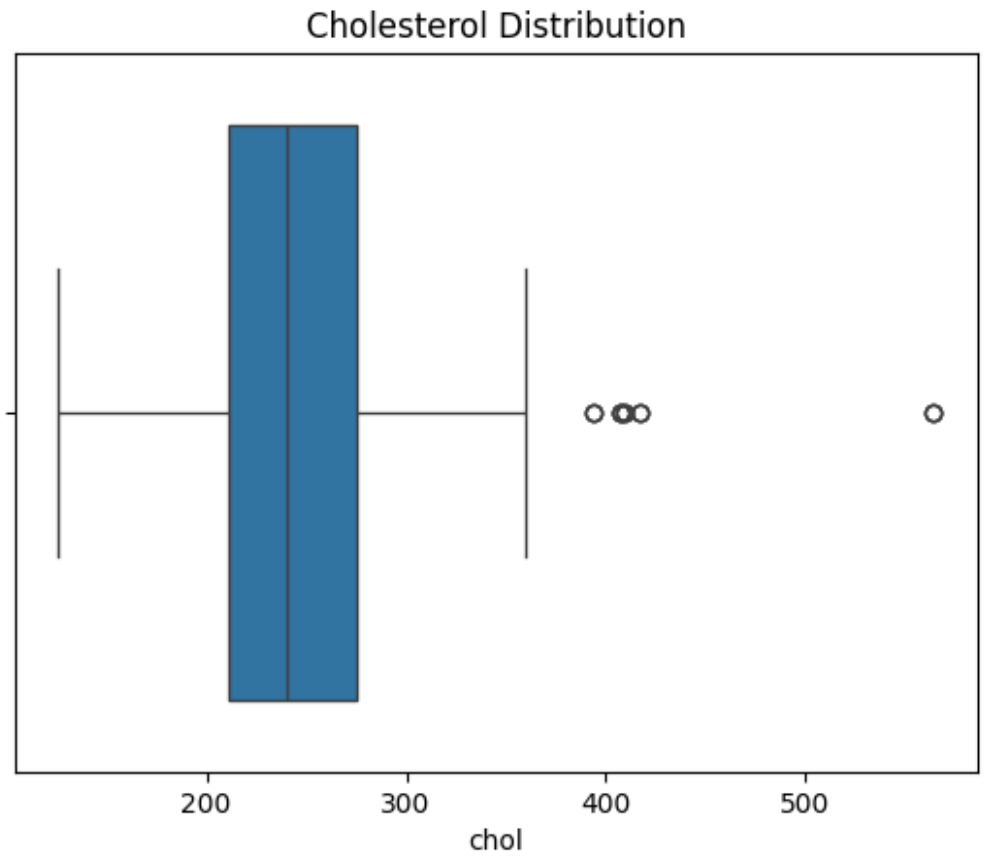
Count Plot

```
sns.countplot(x='target', data=df)
plt.title("Heart Disease Count (0 = No, 1 = Yes)")
plt.show()
```



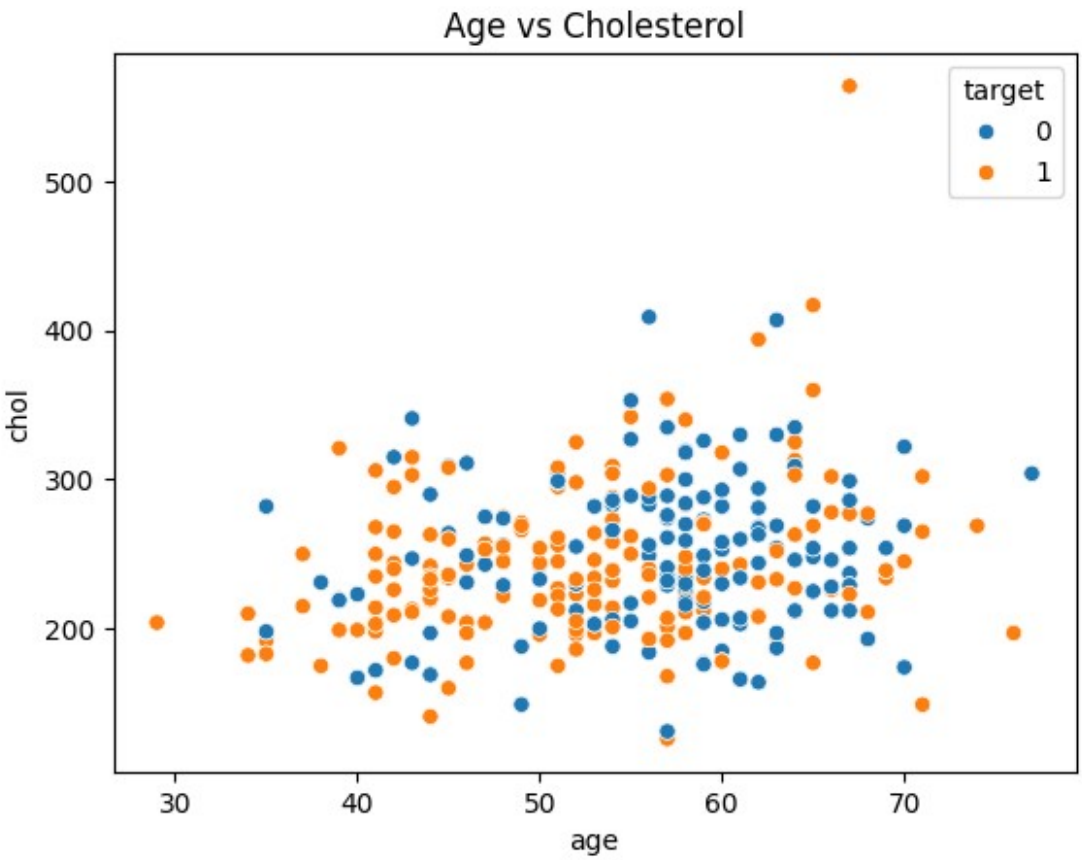
Box Plot

```
sns.boxplot(x=df['chol'])
plt.title("Cholesterol Distribution")
plt.show()
```



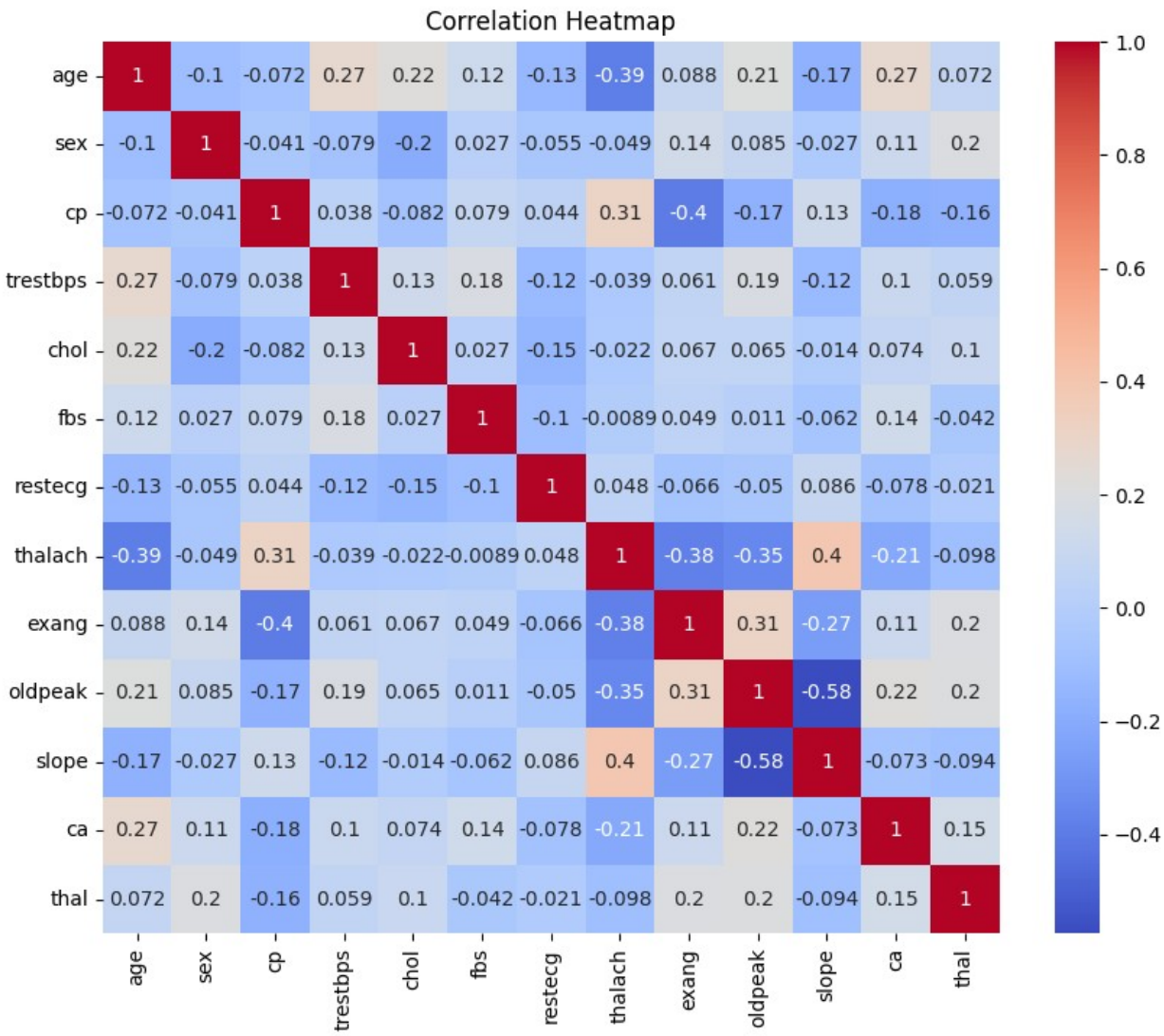
Scatter Plot

```
sns.scatterplot(x='age', y='chol', hue='target', data=df)
plt.title("Age vs Cholesterol")
plt.show()
```



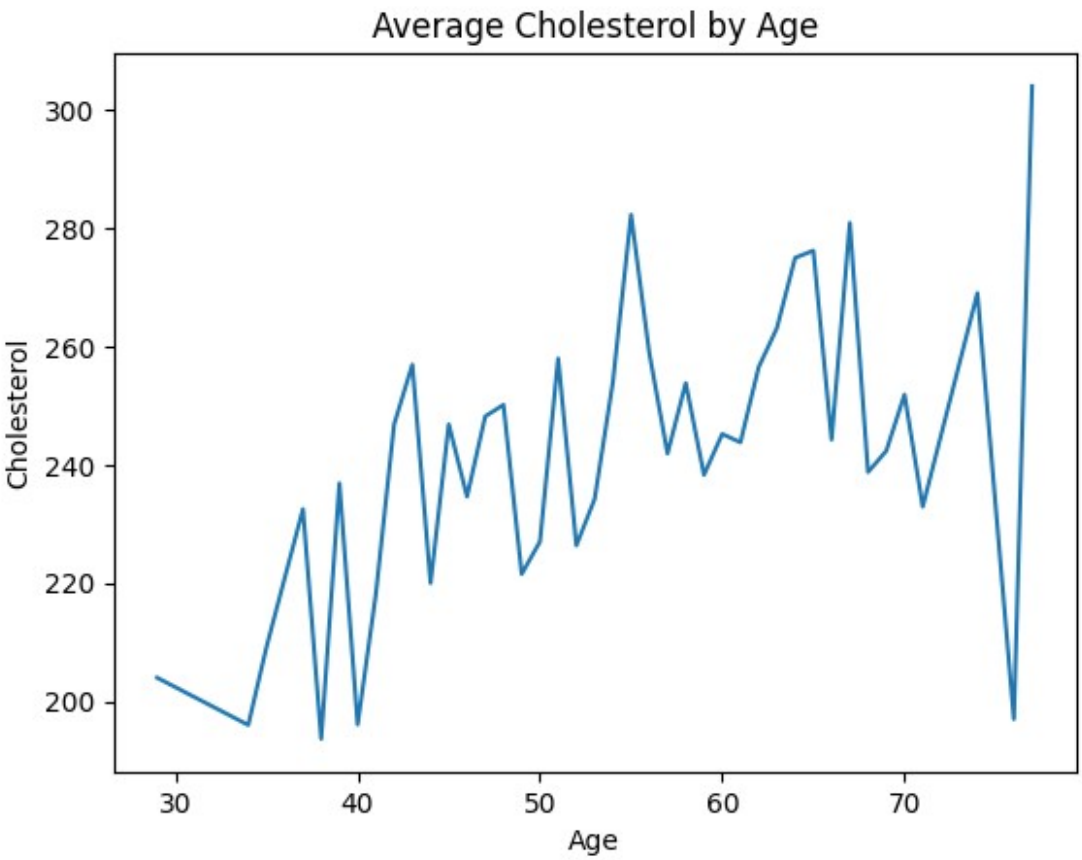
Heatmap

```
plt.figure(figsize=(10,8))
corr = df.corr(numeric_only=True)
sns.heatmap(corr, annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```



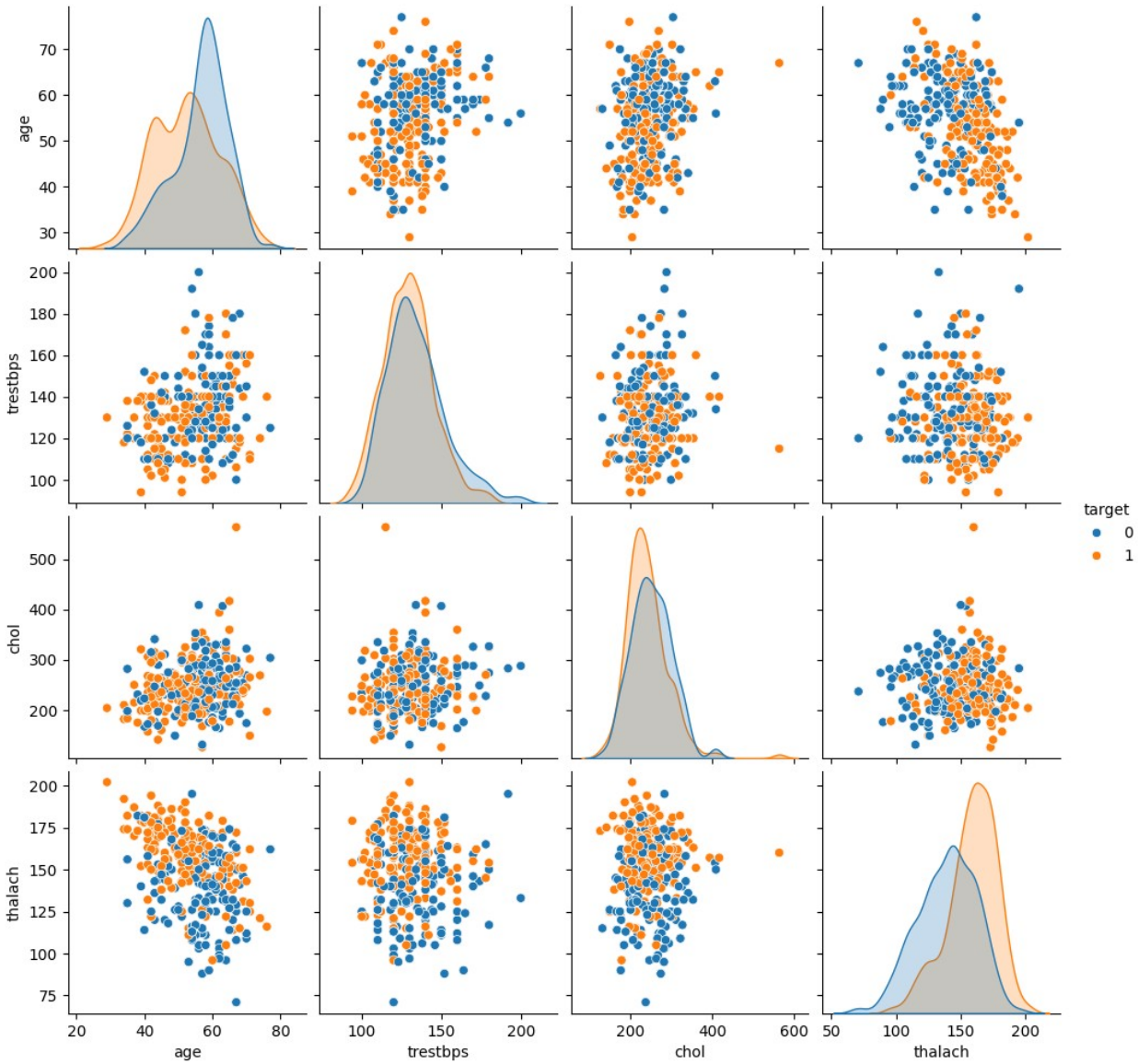
Line Plot

```
df.groupby('age')['chol'].mean().plot()
plt.title("Average Cholesterol by Age")
plt.xlabel("Age")
plt.ylabel("Cholesterol")
plt.show()
```



Pair Plot

```
sns.pairplot(df[['age','trestbps','chol','thalach','target']],
hue='target')
plt.show()
```


age
trestbps
chol
thalach
target
0
1